

Program 1.(a).

1.a) Create the following relation for the student:

Student (regno:string,name: string, class: string, bdate: date, marks1:int, marks1:int, marks2:int, marks3:int)

- i. Enter at least five tuples of the above relation
- ii. Demonstrate the usage of following clauses for the above relation
 - a. Where c. Having
 - b. Order By d. Group By
- iii. Demonstrate the usage of following clauses for the above relation
 - a. Sum c. Count e. Between
 - b. Avg d. Like f. Max & Min
- iv. Demonstrate the rollback and commit command for the above relation

```
mysql> use college14;  
Database changed
```

Create the following relation for the student:

Student (regno:string,name: string, class: string, bdate: date, marks1:int, marks1:int, marks2:int, marks3:int)

```
mysql> create table student(regno int primary key, name varchar(15),  
-> class varchar(5), bdate date, marks1 int, marks2 int, marks3 int);  
Query OK, 0 rows affected (0.06 sec)
```

i. Enter at least five tuples of the above relation

```
mysql> insert into student values(2001,'rakesh','bca4','2004-07-01',70,80,90);  
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into student values(2002,'basu','bca2','2004-10-05',35,40,30);  
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into student values(2003,'chandan','bca4','2004-08-06',70,90,80);  
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into student values(2004,'suresh','bca4','2003-12-11',85,92,96);  
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into student values(2005,'zubair','bca2','2003-05-06',45,50,40);  
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from student;
```

regno	name	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40

```
5 rows in set (0.00 sec)
```

ii. Demonstrate the usage of following clauses for the above relation

- a. Where c. Having
- b. Order By d. Group By

a.where

```
mysql> select * from student where regno=2004;
```

regno	name	class	bdate	marks1	marks2	marks3
2004	suresh	bca4	2003-12-11	85	92	96

1 row in set (0.03 sec)

```
mysql> select * from student where name='basu';
```

regno	name	class	bdate	marks1	marks2	marks3
2002	basu	bca2	2004-10-05	35	40	30

1 row in set (0.00 sec)

b. Order By

```
mysql> select * from student order by name asc;
```

regno	name	class	bdate	marks1	marks2	marks3
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2001	rakesh	bca4	2004-07-01	70	80	90
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40

5 rows in set (0.00 sec)

```
mysql> select * from student order by marks3 desc;
```

regno	name	class	bdate	marks1	marks2	marks3
2004	suresh	bca4	2003-12-11	85	92	96
2001	rakesh	bca4	2004-07-01	70	80	90
2003	chandan	bca4	2004-08-06	70	90	80
2005	zubair	bca2	2003-05-06	45	50	40
2002	basu	bca2	2004-10-05	35	40	30

5 rows in set (0.02 sec)

c. Having

```
mysql> select class, count(*) from student group by class having count(*)>2;
```

```
+-----+-----+
| class | count(*) |
+-----+-----+
| bca4  |      3   |
+-----+-----+
1 row in set (0.00 sec)
```

d. Group By

```
mysql> select class, sum(marks1) from student group by class;
```

```
+-----+-----+
| class | sum(marks1) |
+-----+-----+
| bca2  |      80     |
| bca4  |     225     |
+-----+-----+
2 rows in set (0.03 sec)
```

iii. Demonstrate the usage of following clauses for the above relation

a. Sum

c. Count

e. Between

b. Avg

d. Like

f. Max & Min

```
mysql> select * from student;
```

```
+-----+-----+-----+-----+-----+-----+
| regno | name  | class | bdate  | marks1 | marks2 | marks3 |
+-----+-----+-----+-----+-----+-----+
| 2001  | rakesh | bca4  | 2004-07-01 | 70    | 80     | 90     |
| 2002  | basu   | bca2  | 2004-10-05 | 35    | 40     | 30     |
| 2003  | chandan | bca4  | 2004-08-06 | 70    | 90     | 80     |
| 2004  | suresh | bca4  | 2003-12-11 | 85    | 92     | 96     |
| 2005  | zubair | bca2  | 2003-05-06 | 45    | 50     | 40     |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.06 sec)
```

a. Sum

```
mysql> select sum(marks3) from student;
```

```
+-----+
| sum(marks3) |
+-----+
|      336    |
+-----+
1 row in set (0.00 sec)
```

b. Avg

```
mysql> select avg(marks3) from student;
```

```
+-----+
| avg(marks3) |
+-----+
| 67.2000     |
+-----+
1 row in set (0.05 sec)
```

c. Count

mysql> select count(*) from student;

count(*)
5

1 row in set (0.00 sec)

mysql> select class, count(*) from student group by class;

class	count(*)
bca2	2
bca4	3

2 rows in set (0.00 sec)

d. Like

mysql> select * from student where name like '_a%';

regno	name	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30

2 rows in set (0.00 sec)

e. Between

mysql> select * from student where marks1 between 60 and 70;

regno	name	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2003	chandan	bca4	2004-08-06	70	90	80

2 rows in set (0.06 sec)

f. Max & Min

mysql> select max(marks1), min(marks1) from student;

max(marks1)	min(marks1)
85	35

1 row in set (0.00 sec)

iv. Demonstrate the rollback and commit command for the above relation

```
mysql> set autocommit=off;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> start transaction;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40

5 rows in set (0.00 sec)

```
mysql> insert into student values(2006,'ajay','bca2','2003-04-06',45,89,75);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into student values(2007,'sameer','bca4','2003-10-10',40,52,60);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40
2006	ajay	bca2	2003-04-06	45	89	75
2007	sameer	bca4	2003-10-10	40	52	60

7 rows in set (0.00 sec)

```
mysql> rollback;  
Query OK, 0 rows affected (0.11 sec)
```

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40

5 rows in set (0.00 sec)

```
mysql> savepoint s1;  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> insert into student values(2008,'shashi','bca5','2002-02-11',58,91,87);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40
2008	shashi	bca5	2002-02-11	58	91	87

```
6 rows in set (0.00 sec)
```

```
mysql> rollback to s1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40

```
5 rows in set (0.00 sec)
```

```
mysql> insert into student values(4001,'amit','bba2','2002-05-07',50,55,60);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into student values(4002,'suman','bba2','2002-11-06',60,70,90);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> commit;  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40
4001	amit	bba2	2002-05-07	50	55	60
4002	suman	bba2	2002-11-06	60	70	90

7 rows in set (0.00 sec)

```
mysql> rollback;
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> select * from student;
```

regno	sname	class	bdate	marks1	marks2	marks3
2001	rakesh	bca4	2004-07-01	70	80	90
2002	basu	bca2	2004-10-05	35	40	30
2003	chandan	bca4	2004-08-06	70	90	80
2004	suresh	bca4	2003-12-11	85	92	96
2005	zubair	bca2	2003-05-06	45	50	40
4001	amit	bba2	2002-05-07	50	55	60
4002	suman	bba2	2002-11-06	60	70	90

7 rows in set (0.00 sec)